Binary Trees:

Terms:

1) **Root** - The root, is the ***starting*** position where the first node sits.

2) **Parent** - To easily explain this, ***Root*** = starting $\longrightarrow \forall n, n =$ node that come from the root, are know as ***children*** and root would be the ***Parent***.

3) **Child** - Again, let's make it easy - above I basically explained it… but if you don't understand… the ***child*** in the case about would be the nodes descending from the ***root***.

4) **Sibling** - Those who are ***children*** to the same ***Parent*** are ***Siblings***.

5) **Descendents** - This is hard to explain without a visual but essentially if you pick a node within a tree - it's desencdants are the nodes that are ***Below*** it.

6) **Ancestor** - Ancestors are the nodes that are '***above***' the current node being looked at (which I will elaborate more on)

7) **Degree of node** - The degree is hard to explain without a visual - but the ***Degree*** is essentially the amount of

8) **Internal/External (Terminal/Non-Terminal)** - Nodes that leafs are ***Non-Terminals*** $\forall$ other nodes they are said to be ***Terminals***.

9) **Levels** -

10) **Height** - The height of the tree starting from 0 index, essentially when you draw it out you will assume the ***root*** to be the [0] index.

11) **Forest** - essentially a forrest is when you have a collection of trees.

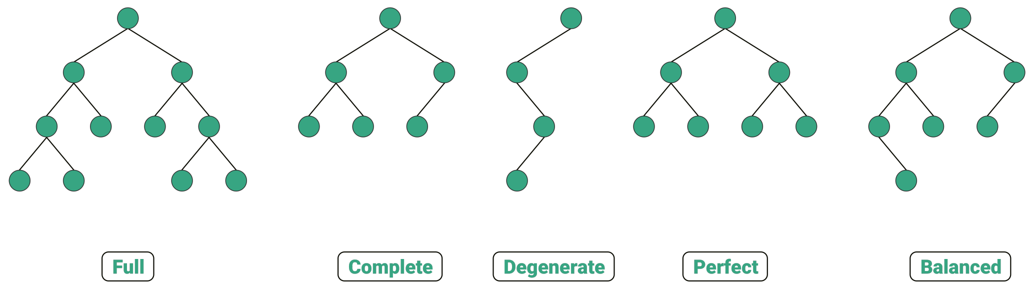

Figure 1: Tree Structures

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1) ***Full*** - A full binary tree is also a ***Strict Binary Tree***!

2) ***Complete*** - A tree that is filled with nodes from ***Left to right*** - or fills by lvl.

3) ***Degenerate*** -

4) ***Perfect*** - A tree that is Perfect $\longrightarrow$ the tree will have the ***Maximum amount of nodes possible*** $\longrightarrow n = 2^{h+1} - 1$

5) ***Balanced*** -

$$T(n) = \frac{(2*n)_c n}{n+1}$$